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Schuckers/Ageenko

DTSC 2302

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Portuguese Language Student Performance Part 2

To build the regression model predicting final grade (G3), we first imported the dataset and removed the variables G1 and G2 to avoid using prior grades as predictors. Categorical variables were converted into dummy variables using one-hot encoding to allow inclusion in the linear regression model. The dataset was then split into training and testing sets (80/20 split) to evaluate out-of-sample performance. A multiple linear regression model was fit using all remaining predictors. Feature significance was assessed using p-values, and multicollinearity was checked using variance inflation factors (VIF). While all predictors were initially included, model refinement focused on retaining variables that were statistically meaningful and contributed to predictive performance. The final model was evaluated using R^2 , adjusted R^2 , and root-mean-squared error (RMSE) on both training and testing datasets.

The prediction model we used was a multiple linear regression model to predict the variable G3, the final exam grade a student scored on. The features used to predict this come from 30 different variables that are based on demographic and family factors and behavioral factors such as school attended, sex, age, parental education, study time, past failures, and absences. Three of the variables that are relevant to this regression are “failures,” which has a coefficient of -1.360 ($p < 0.001$); this means that for every failure that a student has experienced in school, there is a drop of 1.36 points on the final exam grade that is predicted. The highest

positive coefficient of 1.883 ($p < 0.001$); this means that students' academic aspirations are the most important factor in this regression. "Study time has a coefficient of +0.406 ($p = 0.005$).

The model accounts for roughly 35% of the variation in final grades ($R^2 = 0.352$). The F-statistic of 10.3 ($p = 4.6e-37$) shows that the model is better than just using the mean to make predictions. However, the low adjusted R^2 of 0.318n shows that many of the 30 predictors don't add much to the model. Only about 7 of them are significant at the 5% level. Residual analysis also shows some concerns, as the Omnibus test and the residuals show some skewness to the left and heavy tails, which affects the accuracy of the standard errors and p-values. The in-sample RMSE of 2.6 grade points reported for the model is likely to be a very optimistic figure, as the notebook performs no train-test split or cross-validation whatsoever. Until out-of-sample evaluation of the model is performed, its true usefulness remains to be seen, but the model can be seen as descriptively useful in pointing out the importance of aspiration, past failures, and study habits for final grades

Appendix 1

In [36]:

```
# reading in the libraries and functions that we will need as we do this work.
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import scipy.stats as st
import statsmodels.api as sm
import pylab as py

# here are some of the tools we will use for our analyses
from sklearn.linear_model import LinearRegression
from sklearn.metrics import PredictionErrorDisplay
from sklearn.metrics import root_mean_squared_error
from sklearn.metrics import r2_score

from statsmodels.stats.outliers_influence import variance_inflation_factor
```

In [37]:

```
Ames = pd.read_csv("https://webpages.charlotte.edu/mschuck1/classes/DTSC2302/student-performance_n_600.csv", na_values=['NA'])
# remove rows with missing data
Ames.dropna(inplace=True)
Ames.head()
```

Out[37]:

	Unnamed: 0	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	0.648298	MS	F	17	U	GT3	T	1	1	other	...	4	3	2	1	2	5	9	7	9	10
1	0.522011	MS	F	16	R	GT3	T	2	3	at_home	...	4	5	2	1	2	5	0	16	17	17
2	0.119924	MS	M	16	U	LE3	T	2	2	services	...	5	1	3	2	2	3	0	10	9	10
3	0.519012	GP	F	16	U	GT3	T	2	2	other	...	5	4	4	1	1	5	0	12	11	11
4	0.996449	GP	F	16	U	LE3	T	2	4	other	...	4	2	2	1	2	5	2	14	12	13

5 rows x 34 columns

In [38]:

```
from sklearn.preprocessing import LabelEncoder
import pandas as pd
# Create a LabelEncoder object
label_encoder = LabelEncoder()
# List of categorical columns to encode (you can modify this list based on your dataset)
categorical_columns = ['school', 'sex', 'address', 'famsize', 'Pstatus', 'Mjob', 'Fjob', 'reason', 'guardian', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery', 'higher']
# Apply LabelEncoder to each categorical column in the dataset
for column in categorical_columns:
    Ames[column] = label_encoder.fit_transform(Ames[column])
X = Ames[['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu', 'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime', 'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery', 'higher']]
y = Ames['G3']
```

In [39]:

```
# Create a linear regression model
p_model = LinearRegression()

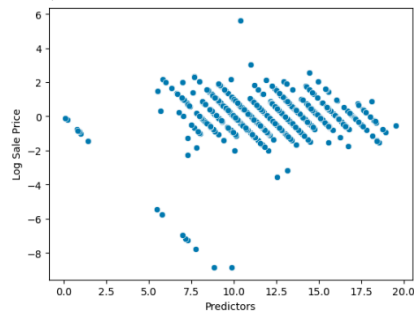
# Fit the model on the data
p_model.fit(X, y)

# Make predictions on the data
y_hat = p_model.predict(X)

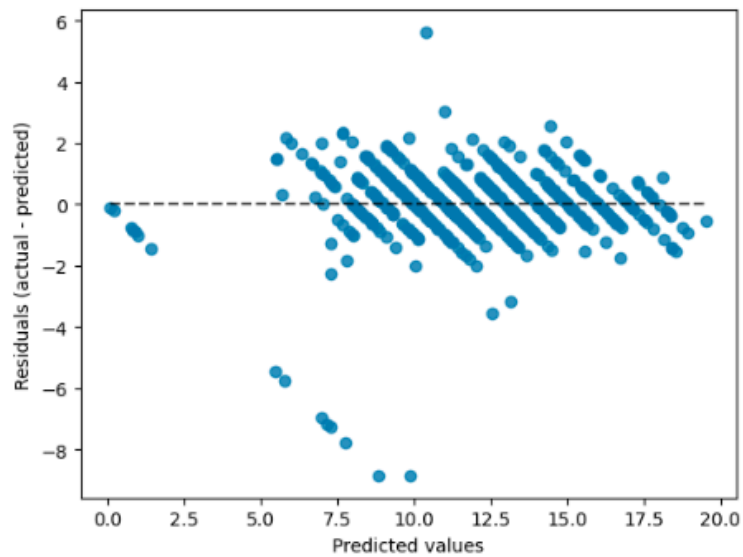
# a look at the residual vs fitted plot
sns.scatterplot(x=y_hat, y=y-y_hat)
plt.ylabel('Log Sale Price')
plt.xlabel('Predictors')
# Evaluate the model performance
mse = root_mean_squared_error(y, y_hat)
print('Root Mean Squared Error:', mse)

# Get the coefficients and intercept
print('Coefficients:', p_model.coef_)
print('Intercept:', p_model.intercept_)

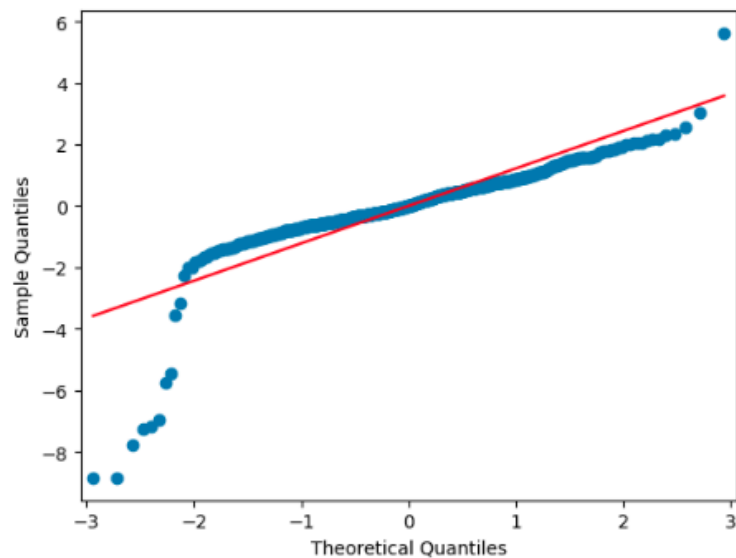
Root Mean Squared Error: 1.2188814317916559
Coefficients: [-0.25318541 -0.08914035  0.05306083  0.10168008  0.02911665 -0.15248818
 -0.04047575  0.03255756  0.01256834 -0.11462626 -0.0888012  -0.01778282
 -0.13537187  0.04588721 -0.21523891 -0.22954185  0.14590674  -0.06194621
  0.04609389 -0.07496312  0.30666642  0.05845565 -0.079181  -0.01139403
 -0.04067104 -0.02734894 -0.02543667 -0.01882778 -0.0537264  0.01227855
  0.13788397  0.06410428]
Intercept: -0.07350388229032667
```



```
In [40]: # below makes a residual vs predicted values plot
display = PredictionErrorDisplay(y_true=y, y_pred=y_hat)
display.plot()
plt.show()
```



```
In [41]: # calculate the residuals
Ames_residuals = y - y_hat
# generate the qq plot and put a line through the points to help us visualize the relationship here
sm.qqplot(Ames_residuals, line='s')
#
py.show()
```



```
In [42]: x2 = sm.add_constant(X)

#fit linear regression model
model2 = sm.OLS(y, x2).fit()

#view model summary
print(model2.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          G3      R-squared:                0.858
Model:                  OLS      Adj. R-squared:            0.850
Method:                 Least Squares      F-statistic:          107.0
Date:                   Tue, 17 Mar 2026    Prob (F-statistic):    3.99e-217
Time:                   16:00:32           Log-Likelihood:       -970.12
No. Observations:      600           AIC:                  2006.
Df Residuals:          567           BIC:                  2151.
Df Model:               32
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	-0.0735	0.972	-0.076	0.940	-1.983	1.836
school	-0.2532	0.129	-1.960	0.051	-0.507	0.001
sex	-0.0891	0.123	-0.725	0.468	-0.330	0.152
age	0.0531	0.050	1.066	0.287	-0.045	0.151
address	0.1017	0.126	0.807	0.420	-0.146	0.349
famsize	0.0291	0.121	0.241	0.809	-0.208	0.266
Pstatus	-0.1525	0.176	-0.865	0.387	-0.499	0.194
Medu	-0.0405	0.068	-0.594	0.553	-0.174	0.093
Fedu	0.0326	0.064	0.508	0.611	-0.093	0.158
Mjob	0.0126	0.049	0.257	0.797	-0.083	0.108
Fjob	-0.1146	0.064	-1.796	0.073	-0.240	0.011
reason	-0.0888	0.045	-1.959	0.051	-0.178	0.000
guardian	-0.0178	0.109	-0.163	0.871	-0.232	0.196
traveltime	0.1354	0.076	1.790	0.074	-0.013	0.284
studytime	0.0459	0.069	0.669	0.504	-0.089	0.181
failures	-0.2152	0.100	-2.145	0.032	-0.412	-0.018
schoolsup	-0.2295	0.176	-1.302	0.194	-0.576	0.117
famsup	0.1459	0.111	1.315	0.189	-0.072	0.364
paid	-0.0619	0.231	-0.268	0.789	-0.516	0.392
activities	0.0461	0.108	0.427	0.670	-0.166	0.258
nursery	-0.0750	0.131	-0.573	0.567	-0.332	0.182
higher	0.3067	0.189	1.626	0.105	-0.064	0.677
internet	0.0585	0.133	0.440	0.660	-0.202	0.319
romantic	-0.0792	0.112	-0.709	0.478	-0.298	0.140
famrel	-0.0114	0.056	-0.204	0.838	-0.121	0.098
freetime	-0.0407	0.056	-0.729	0.466	-0.150	0.069
goout	-0.0273	0.052	-0.528	0.598	-0.129	0.074
Dalc	-0.0254	0.072	-0.352	0.725	-0.168	0.117
Walc	-0.0188	0.057	-0.331	0.741	-0.130	0.093
health	-0.0537	0.038	-1.425	0.155	-0.128	0.020
absences	0.0123	0.013	0.973	0.331	-0.012	0.037
G1	0.1378	0.039	3.545	0.000	0.061	0.214
G2	0.8641	0.036	24.064	0.000	0.794	0.935

```

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Omnibus:                 415.490      Durbin-Watson:           2.106
Prob(Omnibus):            0.000      Jarque-Bera (JB):       8881.538
Skew:                     -2.749      Prob(JB):               0.00
Kurtosis:                 21.029      Cond. No.:              487.
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```

